

keypad made of tactile switches S2 through S9 (Fig. 2). The message appears on the LCD for visual checking. LEDs indicate the applied command. The transmitted Morse code message is played on a piezobuzzer and indicated by the glowing of LED4.

Circuit description

Fig. 2 shows the circuit of the microcontroller-based Morse code encoder. It comprises microcontroller AT89C51 (IC1), regulator 7805 (IC2), and LCD display and a few discrete components. Microcontroller AT89C51 is the heart of the circuit. It is an 8-bit microcontroller with 4kB Flash program-mable and erasable read-only memory (PEROM), 128 bytes of RAM, 32 input/output lines, two 16-bit timers/counters, a five-vector two-level interrupt architecture, a full-duplex serial port, on-chip oscillator and clock circuitry.

Port pins P1.0 through P1.7 of the microcontroller are connected to switches S2 through S9 that are used for inputting the various commands. These port pins are pulled up with 10-kilo-ohm resistor network RNW1. Data pins D0 through D7 of the LCD are connected to port pins P2.0 through P2.7 of the microcontroller. The control pins—register select (RS), read/write (R/W) and enable E—are connected to port pins P0.0, P0.1 and P0.2, respectively. Preset VR1 is connected to pin 3 of the LCD for contrast control.

LED2 through LED5 are connected to pins P0.3 through P0.6 to provide the visual indication for 'EDIT', 'TRANSMIT', 'MORSE OPTICAL' and 'DONE' mode, respectively. Port pin P0.7 of the microcontroller drives the piezobuzzer with the help of transistor T1.

When the buzzer is kept in front of the microphone of a Morse code transmitter, text entered on the display gets transmitted. The Morse code receiver receives the transmitted code and decodes it to retrieve the message. Power-on reset is provided by the combination of resistor R2 and capacitor C3. Switch S1 is used for manual reset. A 12MHz crystal along with two 33pF capacitors

provides the basic clock frequency.

To derive the power supply for the circuit, the 230V AC mains is stepped down by transformer X1 to deliver a secondary output of 9V, 500mA. The transformer output is rectified by a full-wave rectifier comprising diodes D1 through D4. The full-wave rectified output is smoothed by capacitor C1 and regulated by voltage regulator IC 7805 (IC2). Ca-

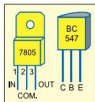


Fig. 3: Pin details of 7805 and BC547

pacitor C2 bypasses the ripples present in the regulated supply. LED1 acts as the power indicator. Resistor R1 limits the current through the LED1.

Fig. 3 shows pin details of voltage regulator IC 7805 and transistor BC547.

Assemble the circuit on a PCB to minimise time and assembly errors. An actual-size, single-side PCB for the microcontroller-based Morse code encoder is shown in Fig. 4 and its component layout in Fig. 5.

Operation

The table shows Morse code of upper-case letters A through Z, numbers 0 through 9 and some special characters.

Power-on the circuit and press switch S1 to reset it. Let's say you want to transmit the message 'HELLO.' This message consists of alphabets, so press key S4 (CH.) to input capital letters from A to Z. Select letters from A to Z by pressing key S2 (UP) for forward movement of the alphabets and key S3 (DN.) for reverse movement of the alphabets. Once the desired alphabet is selected, confirm the selection by pressing 'OK' (S7) key. In a similar way, other

Morse Code of A to Z, 0 to 9 and Some Special Characters

Symbol	Code	Symbol	Code	Symbol	Code
A	.-	Q	--.-	7	
B	-...-	R	.-.-	8	---..
C	-.-.-	S	...-	9	-
D	-..-	T	-	0	-----
E	.	U	..-		
F	..-.-	V	...-	?	...--
G	--.	W	-.-	'	-
H		X	-.-	.	...--
I	..	Y	..-	-	-
J	...-	Z	--.-	"	...--
K	-.-	1	-	(	...--
L	..-.	2	-	)	...--
M	--	3	-		...--
N	-.	4	-		
O	---	5	-		
P	.-.	6	-		

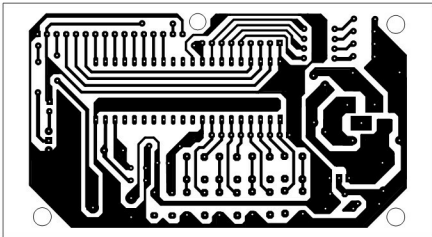


Fig. 4: An actual-size, single-side PCB for the microcontroller-based Morse code encoder